Parallel circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate branch currents to a shared potential difference.

SUCCESS CRITERIA - I can...

1. Identify branches connected across the same two nodes.
2. Use equal voltage across parallel branches.
3. Add branch currents to find supply current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Electrons in a wire drift at 0.3 mm/s while a lamp responds quickly after a switch closes. Explain why response time cannot be found by timing one electron across the wire.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $1/R_{total} = 1/R_1 + 1/R_2$; $I_{total} = I_1 + I_2$; $V_1 = V_2$. Parallel branches connect to the same two nodes and share voltage.

Worked reasoning: Two parallel branches of 6 and 3 ohms each have 6 V across them. Branch currents are 1 and 2 A; source current is 3 A.

Your turn: Draw the two branches and mark where the currents split and recombine.

Marzano check - circle: ① ② ③

03 Represent and interpret

Mark the two shared nodes in the parallel diagram. If one branch opens, can the other remain powered? Explain.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Solve and justify

A 12 V source powers 6 Ω and 12 Ω in parallel. Find branch currents, supply current and equivalent resistance.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Why can the equivalent resistance be smaller than either branch resistance?

Marzano check - circle: 1 2 3

06 Choose a method

A converging lens has f=20 cm and d_o=8 cm. Use $1/f=1/d_o+1/d_i$ and $m=-d_i/d_o$. Find d_i, magnification and image kind.

Marzano check - circle: 1 2 3

07 Exit check

Two equal 8 Ω resistors are in parallel. Find equivalent resistance. Compare with the same resistors in series.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 12 V supply feeds parallel 4 and 6 ohm resistors. Find equivalent resistance, total current and total power by two methods.

Marzano check - circle: 1 2 3